A National Airspace System Reference Architecture as a System of Systems

Tracing Operational Intent from Enterprise Objectives to Aircraft Trajectory

ISD 503 Capstone Project Report – Interim Report #1

Submitted to:
Mark Petrotta - Principal Engineer of Digital Engineering, MBSE, and Augmented Intelligence - Aerospace and Defense Systems
IOE Department of Integrated Systems Engineering & Design

Prepared by:
Owen O'Keefe
University of Michigan, ISD

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Plan ref: to-do-list.md S1 — Confirm advisor/reader name and title for "Submitted to."

Executive Summary

Plan ref: to-do-list.md S14, S16 — Write this section LAST. Distill: (1) the improvement opportunity/problem statement, (2) the MBSE approach used, (3) major architecture and optimization-demonstration findings stated quantitatively, (4) the operational impact/benefit of the reference architecture (see Section 5.0 Impact/Financial Benefits), and (5) recommendations. Limit to 1-2 pages; no citations, undefined terms, or content not found elsewhere in the report.

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1 Introduction and Current State

REQUIRED FOR INTERIM REPORT #2 (due 2026-10-25) — Introduction. Interim Report #2 resubmits this section updated. Three edits are needed: (1) the "current state of the project" paragraphs in Section 1.2 still describe mid-September ("off-track, but recoverable", state machines due 9/15) and should describe October: the experiment defined (D-013 to D-015), the plan re-based on four tracks and nine turns (D-016), the scoring model chosen (D-017); (2) the research-gap placeholder below needs the owner's paragraph; (3) the project timing subsection should show the nine turns, not only the August table.

Plan ref: to-do-list.md S1 — Briefly state the subject, purpose, and scope of the report: the NAS modeled as a System of Systems, centered on how operational intent propagates from enterprise objectives to aircraft trajectory/behavior.

My project aims to build a reference MBSE architecture for how airspace intent is managed across the NAS — spanning the organizations that define and enforce rules, the domains managing airspace resources, the systems supporting flight operations, and the onboard systems that turn intent into executable trajectory and guidance behavior.

In his 2005 paper, DeLaurentis states the following DeLaurentis [16]:

> "Problems of significant complexity are facing decision-makers within government and industry, and the moniker of 'system-of-systems' (heretofore shortened to SoS) is increasingly being applied to problems of this type. Effective system analysis for decision-support quickly becomes unmanageable in this context, with multiple, heterogeneous, distributed systems involved and couched in networks at multiple levels. The situation is exacerbated by the fact that most organizations (and their processes) remain in the 'stovepipe' mode – their people and tools are configured to study within narrow bins with little or no analysis across those bins. Current frames of reference, thought processes, analysis, and design methods are not complete for these SoS problems ... A holistic framework is needed that enables decision makers to discern whether related infrastructure, policy, and/or technology considerations together are good, bad (or indifferent) over time."

While this reference is from 2005, it is still relevant today. The NAS is a complex system of systems, and the decisions made by the various stakeholders have significant consequences for the safety, efficiency, and sustainability of the air transportation system.

Airline companies invest significant time and money in constructing operational plans that maximize revenue; however, on the day of operation, unpredictable events – bad weather, aircraft malfunctions, crew absenteeism – can substantially alter that plan and the revenue objectives behind it Castro [11]. To manage this, airlines maintain an Airline Operations Control Center (AOCC), a department of experts specialized in disruption management across aircraft, crew, and passenger recovery. The AOCC monitors operations and makes real-time decisions – re-routing flights, rescheduling crews, rebooking passengers – intended to minimize the impact of a disruption on the airline's schedule and revenue objectives.

The literature does not converge on a single, standardized definition of the "cost" the AOCC is minimizing, and the values it does quantify do not always hold up under scrutiny. Delay cost is commonly modeled in the optimization literature as a linear function of delay minutes, but Hu et al. [44] notes that this assumption is inconsistent with airline practice, where delay cost is in fact a nonlinear function of delay time – and that, faced with this complexity, airlines tend to prioritize the speed of a recovery decision over the mathematical optimality of the resulting solution. This preference for tractable, fast decisions over globally optimal ones shows up structurally as well as behaviorally: a review of the Aircraft Recovery Problem finds that studies – and, by

extension, the AOCC processes they model – rarely minimize delay, cancellations, and aircraft swaps simultaneously, instead treating aircraft, crew, and passenger recovery as separable subproblems solved sequentially and handed off in turn Santana et al. [76]. Each stage can produce a solution that is locally optimal yet globally suboptimal, or even infeasible, once the next stage's constraints are applied – a concrete, literature-documented instance of the myopic-optimization pattern this capstone investigates.

That pattern's costs and benefits are visible even in work that improves on the status quo. Kontodimou et al. [56]'s predictive turnaround-buffer allocation model achieves a 37.1% reduction in propagated delay relative to a zero-buffer baseline,but it reaches that result by holding aircraft routing, fleet assignment, crew duty, maintenance, and passenger connectivity fixed as upstream constraints, deliberately excluding them from the optimization it performs. The gain is real within that bounded scope, but the framing illustrates the exact risk this capstone is concerned with: a locally-optimized decision layer can report a strong local benefit while externalizing tradeoffs onto adjacent decision layers that go unmeasured by that layer's own cost model. The same pattern appears on the human side of the AOCC: Turhan, Gzara, and Elhedhli [88] documents that when dispatcher workload exceeds capacity, dispatchers "may resort to conservative approaches, which can have financial implications for airlines" – a workload-driven cost that rarely appears in the delay, cancellation, and swap cost models discussed above. Hassan, Santos, and Vink [36] frames this more broadly as a persistent gap between the disruption-management optimization literature and the reality of AOCC practice.

In this paper, I explore the value tradeoffs that aviation stakeholders experience when disruptions occur and recovery solutions are employed with various levels of myopia. When a dispatcher or AOCC team member makes a decision, they are often focused on the immediate impact of that decision – e.g., minimizing delay for a specific flight – without visibility into the broader implications for the airline's overall operations or the NAS as a whole. Cost models used today typically capture the factors that are easiest to quantify, such as fuel consumption and crew costs, while factors that are harder to quantify – maintenance cost, passenger satisfaction, dispatcher/controller workload, environmental impact – are more often overlooked. This can lead to suboptimal outcomes: increased congestion in certain airspace sectors, missed opportunities for more efficient routing, or decreased overall profits driven by customer dissatisfaction. By modeling the NAS as a system of systems and analyzing the propagation of operational intent, I aim to identify strategies for improving decision-making and reducing the negative impacts of disruptions on the air transportation system.

1.1 Overview of the Project

Plan ref: to-do-list.md S1 — Describe the industry/environment (commercial aviation and the NAS), the social context, and the key stakeholders (preview – full analysis in Section 1.3). State the project's research questions from Project_To-Do List.md S1 ("Finalize project research questions") and explain why they matter. Draw on README.md's framing of the trajectory-intent lifecycle (Strategic objective -¿ mission plan -¿ flight plan -¿ ATC constraints -¿ trajectory negotiation -¿ FMS intent -¿ guidance commands -¿ aircraft motion) as the throughline.

The NAS is a complex system in which operational decisions are made by independent actors, each of which may or may not be accounting for the impacts on other system stakeholders, which includes regulatory agencies, air traffic service providers, airport operators, airline operations centers, MRO suppliers, flight crews, passengers, and aircraft systems. While substantial architectural work has been performed within individual domains, the relationships and information flows between these domains are often incompletely represented.

Existing models frequently focus myopically on one actor or problem, exhibiting one or more of the following limitations: - Emphasis on enterprise-level capabilities without sufficient representation of aircraft behavior and onboard desision-making, - Emphasis on aircraft systems without adequate representation of the broader operational context within which those systems operate, - limited traceability between strategic objectives, operational constraints, human decisions, and the resulting aircraft trajectories, - difficulty evaluating the impacts of new operatioal concepts, automation technologies, and optimization services across organizational boundaries.

As a result, there is no broadly applicable architecture that fully captures how strategic objectives become executable aircraft behavior. This lack of end-to-end traceability limits the ability of engineers, operators, and decision makers to understand the consequences of changes within the NAS and complicates the integration of future decision-support technologies.

This project seeks to address this gap by developing a System-of-Systems architecture that explicitly models the propagation of intent throughout the National Airspace System, from enterprise objectives and airspace management decisions to flight deck decision-making and aircraft trajectory execution.

1.2 Review of Current State Design / Literature Review

Plan ref: to-do-list.md S2–S4 — Synthesize (do not merely list) findings from the three literature-review tracks: airline planning and operations (schedule planning, fleet assignment, crew scheduling, maintenance routing), day-of- operations and turnaround (A-CDM, weight and balance), and OCC/dispatch and flight execution (disruption management, dispatcher responsibilities, nominal ATC/IFR flow). Pull from per-source annotations captured via the literature-review workflow into `evidence/source-register.md` rather than re-deriving them here. Identify the research gap this capstone addresses (see also Project_To-Do List.md S16 "Research gap").

Plan ref: to-do-list.md Track C (research gap) — What this project adds that prior work does not. The opportunity itself is documented: [15] measured what airline-supplied weight does for climb prediction; SESAR's ADS-C EPP downlinks the FMS's predicted profile; [19] gives the aircraft a picture of the traffic. State the three things this project does that those do not: (1) scores one decision at four stakeholder levels side by side (aircraft, airline, sector, system) with no weighted sum, so who pays is visible; (2) maps the decision to its authority and its information owner in a system-of-systems model, so the missing exchange is a named interface (`AirlinePerformanceShare`, D-019) and an unmet requirement (SLR-INF-02, D-018), not a finding in prose; (3) swaps the scoring model's fidelity under a fixed architecture (D-017), which is the MBSE claim.

Plan ref: to-do-list.md S11 (context for the experiment) — What the sector controller knows when resolving a conflict. Make the point that the controller has type, level, speed and route, but not weight or cost index. Cite [78] (p. 2, weight and speed intent "are considered competitive parameters by airlines and are not available for use") and [15] (one nominal weight per aircraft type). Say who does hold the weight (airline, FMS) and that they do not hold the authority.

Plan ref: to-do-list.md S11 (context) — What controllers do by default. Cite [72] (23 of 36 level-crossing cases resolved by a descent; controllers "use more predetermined rules"), [55] (leave the aircraft in steady state alone) and [34] (instruction counts per heuristic; "No point asking aircraft to climb if they come back and say they can't"). Note the disagreement between the first and third on whether descent is preferred.

Plan ref: to-do-list.md S11 (context) — TASAR as the mirror-image study. Cite [19]. Three points to make: (1) it gives the aircraft a picture of the traffic, where this project gives ATC a picture of the aircraft; (2) size of the effect per flight (27 gallons and 2.3 minutes for the typical case, range 12 to 103 gallons, p. 12); (3) it did not model aircraft weight and limited climbs to FL350 and below for that reason (p. 8). Point (3) is the assumption Section 2.4 adopts.

Plan ref: to-do-list.md S11 (context) — How sector loading reaches the controller. Explain the Monitor Alert Parameter in one or two sentences and what a "red sector" is. Cite [28] (§18-9) for the definition and [19] (p. 10) for the practice: the sector controller is generally not aware of red sectors elsewhere, and a supervisor may restrict traffic into one. Open: the report's "area manager" is not the order's term; red versus yellow is not defined in the order.

Plan ref: to-do-list.md S11 (context) – SOURCES NOT YET HELD — Aviation emissions and what they cost whom. Needed before Sections 3 and 5 can discuss emissions: (a) the CO2 produced per kg of jet fuel, with a citation; (b) whether any carbon-pricing or offsetting scheme applies to a U.S. domestic flight, and a dated price; (c) a study of how emissions labels on flight listings affect passenger choice; (d) a social cost of carbon for the system-level discussion. None is in `evidence/sources/` as of 2026-10-09.

From my perspective, the current state of the project is off-track, but recoverable. The plan calls for 4 phases of work, with phase 1 completing 9/10 with RACII matrices for the various objectives within the airspace. More time has been spent on the literature review than originally expected, as the scope broadened to encompass consideration of which ontological model to use. Moreover, a lack of discipline on extracting specific actionable information from the review has weakened definitive claims about specific objectives pursued by stakeholders, which in turn feed the considerations of the RACII matrix. While much learning has occurred from the literature review, the impact on schedule continues. Presently, 53 papers have been extracted, with more than half manually reviewed, with many annotated. The literature review is ongoing, but the current content supports the research claims and the enforces the value of the project deliverables.

Time spent developing the Concepts of Operations has also impacted other work. However, the project lacks a specific focus at the moment for what tradeoffs will be considered. While the project does not intend to be an exploration of optimization, the concept focuses on sub-optimality driven by myopic decisions. While the concepts of operations help inform a more comprehensive understanding of the NAS, the systems involved, and the interfaces amongst them, it has not yet provided for unique insights into specific scenarios that might be worth modelling.

At this stage, the progress is at risk of analysis paralysis, which is being weighed against the risk of a lack of a well-formed vision for execution. At this point, I wish to heir on the side of action, transitioning the project to Phase 2 and allowing specific gaps or needs in the model development guide further review of the collected sources. That said, I still plan to contiue the review ad-hoc. Also worth noting is that model development will inform possibilities w.r.t. what can be modelled (e.g. multiple instances of an aircraft system performing similar, but different actions simultaneously while each affecting the same controller's workload). Additionally, gaining experience with sysmlv2 will inform whether its learning curve makes it a viable tool, or if Cameo is more appropriate.

This pivot from Phase 1 exploration to Phase 2 modelling will advance the project deliverables while engraining a deeper understanding of the information gaps. When the modelling gets to the point of MOEs/MOPs, the objectives of stakeholders will be revisited and a more targetted literature consultation can occur.

Per the original plan, state machines for the various NAS systems were intended to be completed by 9/15. This deliverable has not been met. The realities of a full-time professional job whose needs often extend beyond the normal 40-hour work week has forced much of the project work to occur on weekends and my time managment needs to adjust to accomodate. This weekend will be a decision gate that informs that process and possibly lead to an adjustment of the schedule of deliverables.

(Section 1.5).

1.3 Stakeholders

Plan ref: to-do-list.md S7 — Identify key project stakeholders (individuals/organizations affected by or perceiving themselves affected by NAS operational decisions). Draw on the PESTLE stakeholder discovery, stakeholder register, and RACCI matrix from Project_To-Do List.md S7. Confirm the stakeholder set is diverse (airline, passenger, ATC/ANSP, airport, flight crew, environmental/ societal, military – see S8's per-stakeholder objective categories) and that stakeholders with little formal power (e.g., passengers, communities affected by noise/emissions) are represented.

I created the extttcapstone extunderscore stakeholder extunderscore assessment.xlsx file, filling it with my initial assessment of each candidate stakeholder's objectives, and then iteratively refined it as I read more literature and gained a better understanding of the NAS. The assessment includes the following information for each stakeholder:

- stated interest and whether it acts as a hard constraint, optimization objective, cost/penalty, MOP, or MOE
- candidate explorations of MOP/MOEs
- determination of whether the stakeholder has a direct or indirect impact on trajectory decisions, if any.
- An exploration of whether the stakeholder should be a standalone stakeholder or consolidated with another stakeholder, and the rationale for that decision.

This has lead me to the following list of stakeholders that I believe are relevant to the project:

- Governments / Standards Organizations
- Operators (Airlines, Charter Operators, Cargo Operators)
- Air Traffic Control (ATC)
- Aircraft and Engine Manufacturers
- Airport Authority
- Flight Crew
- Passengers
- Environmental/Societal Organizations

Furthermore, an exploration of the interests of each stakeholder has lead me to the following list of objectives that I believe are relevant to the project:

- Safety / Security
- Efficiency
- Throughput / Capacity
- Reliability / Resilience
- Environmental Impact
- Noise Impact
- Passenger Satisfaction
- Schedule Reliability
- Cost / Profitability
- Maintenance / Lifecycle Cost
- Regulatory Compliance
- Workload / Human Factors

1.4 Scope

Plan ref: to-do-list.md S1 — State what is explicitly included and excluded from the System of Interest (SOI), the level(s) of abstraction modeled, and the criteria used to draw system and stakeholder/actor boundaries. This section should resolve the open boundary questions currently tracked in `knowledge/questions/open-questions.md` S1 – if any remain open when this section is drafted, that is a signal the SOI work in Project_To-Do List.md S1 is not yet closed.

See Section 1.1 for a discussion of the scope of the project. Also note ambiguities highlighted in the review of the current state design/literature review (Section 1.2) that will be addressed in future phases of the project.

1.5 Project Deliverables

Plan ref: to-do-list.md S10, S14, S16 — State the goals and deliverables: the SysML/Cameo reference architecture, the traceability chain from stakeholder needs to aircraft-level behavior, and the optimization-demonstration capability (Section 3). Include the quantifiable measures used to assess success (e.g., traceability coverage, stakeholder coverage, demonstration of a local-vs-SoS optimization tradeoff).

The following deliverables were established for the project upon its inception:

See the review of the current state design/literature review (Section 1.2) for a discussion of the current status of these deliverables and the risks to their timely completion.

1.6 Project Timing

Plan ref: to-do-list.md S1 (new) — Revisit the timeline from the 503 Proposal submittal (see `prework/ISD 503 Submittal.pdf`) and list the sequence of tasks – keyed to Project_To-Do List.md's 16 sections – with target completion dates. A "Project Timing" task did not previously exist in the plan; it has been added to Project_To-Do List.md S1 this session so this subsection has something concrete to draw from.

2 Assumptions and Methodology

The problem is: the inertia of past, piecemeal designs and the high cost of upgrading the NAS make it hard to change the system. The project proposes a new airline-to-ATC interface and a new way of deciding which aircraft to move in a conflict, that together have the ability to lower the cost of air traffic management.

Towards those ends, the projet employs a model-based systems engineering (MBSE) approach to design the architecture and the decision-making process, and a simulation-based experiment to test the proposed interface and decision-making process. First, a literature review was conducted to:

1. Identify the current state of air traffic management,
2. Identify the relevant stakeholders and their objectives,
3. Identify the requirements that are necessary to meet those objectives, and
4. Identify the gaps in the current system that the proposed interface and decision-making process can address.

Table 1: Project milestones and verification methods

Project Milestone	**Deliverable**	**Lead/Team Member**	**Expected Completion Date**	**Target Metric/Verification Method**
Literature Review	Annotated bibliography	Owen	9/7/2026	10–20 papers categorized
PESTLE Analysis & Systems-of-Interest Definition	Context diagram, text-based documentation	Owen	9/10/2026	Review
Systems-of-Interest Definition	Text-based documentation	Owen	9/15/2026	Clear boundaries and exclusions
Operational Scenarios	ConOps document	Owen	9/22/2026	1–3 representative scenarios
Draft Architecture	BDDs and package structure	Owen	10/1/2026	All domains represented
Interface Definition	Interface Control Document	Owen	10/12/2026	All interfaces identified
Internal Block Diagrams	IBDs	Owen	10/17/2026	Critical interfaces connected
Requirements Definition	Requirements model	Owen	10/20/2026	Trace to stakeholders
Measures of Effectiveness	MOE/MOP hierarchy	Owen	10/25/2026	Quantifiable metrics defined
Requirement Traceability	Requirements matrix	Owen	10/29/2026	Complete allocation
Activity Diagrams	Operational activities	Owen	11/3/2026	Scenario walkthrough
Sequence Diagrams	Information exchanges	Owen	11/8/2026	Message completeness
State Machines	State behavior of key entities	Owen	11/13/2026	Executable state logic
Authority & Responsibility Models	RACII (Responsible, Accountable, Consulted, Informed)	Owen	11/17/2026	All decisions assigned
Optimization Prototype	Working code	Owen	11/21/2026	Unit test cases
Integration into MBSE Architecture	Traceability links	Owen	11/24/2026	End-to-end scenario
Final Baseline & Final Paper	Versioned model & thesis/paper	Owen	12/16/2026	Draft
Paper Submittal	Thesis/paper	Owen	12/20/2026	Official submittal

2.1 Systems Engineering Approach

Standard systems engineering methods of systems analysis and decomposition were used on the collected evidence, including generating a context diagram, internal block diagrams, and activity and sequence views. These methods were used to identify the relevant systems and the interactions amongst them that are necessary to meet the stakeholders' objectives. These were used to generate the stakeholder requirements that are necessary to meet those objectives, which were then decomposed into NAS systems-level requirements. The NAS SLRs were then allocated to the relevant sub-systems and HLRs decomposed those one level further.

This enabled the project to: (1) build a model of the architecture and the decision-making process, (2) Compare the current air traffic management practice, which leverages general rules of thumb and knowledge of successful historical actions [**ref:TBD˙TODO**], with a model-based decision support approach that replaces the low-fidelity decision rule with one or more higher-fidelity models of aircraft performance data of varying levels of accuracy.

The working assumptions can be categorized in two ways: (1) assumptions about the problem and its context, and (2) assumptions about the methodology. The problem statement assumes that:

- the abstractions used in the model are sufficient to capture the relevant behavior of the system
- the stakeholders' objectives are comprehensively captured and accurately represented in the model

Every abstraction was done with the expressed goal of capturing the relevant behavior of the system, as defined by the stakeholders' objectives. The decision was made to ratify the NAS-as-SoS System of Interest around the trajectory-intent lifecycle from enterprise objectives through aircraft motion. An entity is modeled when it has distinct decision authority or execution behavior dynamically coupled to in-scope decisions and materially affects cost, safety, workload, schedule reliability, or passenger value. Boundary actors supply objectives, demand, feedback, or rules without being decomposed as decision mechanisms. Exogenous influences enter as constraints or scenario parameters. This criterion preserves the authority and responsibility relationships needed for a systems-of-systems architecture without expanding the project into a model of every NAS participant.

The methodology does not assume that the proposed methods are better, which is tested as part of the experiment. Instead, it assumes that:

- models of every required aircraft type are available
- the security concerns of sharing airline data can be mitigated
- the proposed interface can be implemented in the NAS
- digital communication channels exist between the airline and ATC
- the specific scenario presented in the experiment is representative of a larger set of scenarios which generalize to the NAS as a whole

An emphasis is put on structure/behavior/interfaces/traceability rather than optimization math, as the project's commercial value for the author lies in it furthering their understanding of the systems, their interactions, and the complex series of responsibility divisions that inhibit information flow and lead to sub-optimal outcomes. In order to explore this concept in a meaningful way,

decision-support/optimization is treated as a capability inside the architecture, which is enabled by following MBSE principles.

While more detailed documentations of the systems and their constituent systems are possible (with examples contained in [**appendix˙what˙TODO**]), and more formal mathematical models can be employed, such an effort was deemed outside the scope of the work required to achieve the commercial value for the customer. In true customer-focused MBSE fasion, the project's artifacts are tailored to the customer's needs.

As such, individual steps in the process were tailored to the availability of the author, including a continuous, ad-hoc review of relevant literature, after an initial review was conducted. A full gate-to-gate conops was unofficially constructed and refined with publicly available general information/knowledge of the NAS. No alternative system decompositions were formall explored past initial abstraction decisiosn. In this way, the ISO/IEC/IEEE 15288 technical processes were followed, which includes tailoring them to the application.

The following table summarizes the systems engineering processes and the artifacts that realize them.

Table 2: Systems engineering processes and the artifacts that realize them.

Process (15288 / INCOSE)	Artifact in this project	Where
Stakeholder needs and requirements	27 stakeholder requirements, each traced to an objective row	`requirements_stakeholders.sysml`; Appendix
System requirements	53 system requirements, baseline or improved (D-018), allocated to constituent systems	`requirements_nas_system.sysml`
Architecture definition	Context diagram, internal block diagrams, typed interfaces on the thread (D-019), activity and sequence views	`cameo_models/`; Section 3
System analysis	Experiment at three fidelities (D-017), outcome table	`simulation/`; Section 3.1
Verification	Verification cases for the improved requirements, run by the experiment	`nas_verification.sysml`
Validation	Stakeholder-level reading of the outcome table (who pays)	Section 3
Technical measurement	MoEs (outcome-table columns), MoPs (requirement measures)	Section 2.4
Configuration management	Decisions log, model text as the authoritative source (D-006), model freeze	`decisions-log.md`
Tailoring	D-016	this subsection

Plan ref: to-do-list.md S6 — Decomposition kept, alternatives named. One paragraph: why the authority / responsibility / information-ownership decomposition (D-002, D-007) was kept, naming the alternatives (organization-based, lifecycle-based, physical, information-flow, decision-authority) and the criterion (which one makes the decider, the information owner and the authority visible on the same picture). Building the alternatives is future work (D-016).

2.2 The Two Claims

Plan ref: to-do-list.md Track C (D-016, D-017) — Separate the two claims and say which the numbers support. Claim 1, tested by the experiment: replacing the decider's low-fidelity rule of thumb with a higher-fidelity model fed by weight and cost index across the airline-to-ATC interface changes which aircraft is moved and lowers total system cost (whether one extra clearance is worth it is a result, not an assumption). Claim 2, shown by the model: the architecture locates that interface, names who owns the decision and the data, states the unmet requirement, and lets the scoring model's fidelity be swapped (conceptual, nominal mass, actual mass) with the structure unchanged. The simulation would give the same numbers without the model; say so, and say what the model contributes that the numbers do not.

2.3 Proposed Airline-to-ATC Interface

Plan ref: to-do-list.md Track C (D-019) — Name the digital-thread artifact. Describe `AirlinePerformanceShare`: what it carries (gross weight, cost index, connection priority; thrust, drag and fuel-flow model identifiers present but not exercised), who sends it (airline operations center or FMS), who receives it (ATC decision support, per D-012 owned by the system that decides), and which requirements it satisfies (SLR-INF-02) and the goal it serves (SLR-CLR-13). Position it against ADS-C EPP and Coppenbarger's Table 2 (the same items, a different receiver). State that security is an assumption.

Plan ref: to-do-list.md S10 — Describe the MBSE methodology used to build the architecture in SysML v2 (textual notation authored in Syside Modeler in VS Code, with diagrams as generated views): package organization, view types produced (BDD, IBD, activity, sequence, requirements), and the traceability approach (stakeholder needs -¿ objectives -¿ requirements -¿ systems -¿ activities -¿ information exchanges -¿ aircraft behavior). Note per `workflows/update-architecture.md` (decision D-006) that the SysML v2 text is the source of truth; Cameo is not used as the source of truth. Describe the working loop (draft, print, hand review, transcribe, apply, log, re-render) and the artifact states 0 to 4 (D-016).

2.4 Experiment Design

Plan ref: to-do-list.md S11 — State the central claim the experiment supports. Still undecided (see `experiment-definition.md` §1 and the hand review notes 1.2, 2.1, 4.1). Answer the objection that the simulation compares two ways of operating and so does not by itself test MBSE.

Plan ref: to-do-list.md S11 — Scenario. Four aircraft in one invented en-route sector, shown as one snapshot: a crossing pair at the same level and two aircraft in trail 2,000 ft below. State which aircraft belong to the same airline, which is late with connections, and the weight of each. Give the level of the crossing pair and say why it is above FL350. Justify the static snapshot by the controller studies that used one ([55], [34]).

Plan ref: to-do-list.md S11 — The one thing that differs between cases. Define cases A, B and C by what the decider knows. State what "automated decision" means for who devises and who issues the clearance (undecided). State that the levers are the same in all three.

Table 3: PLACEHOLDER – the three cases. Wording to be the author's.

Case	Decider	Information	Climb rule applies?
A	Controller, by rule	Type, level, speed, route	Yes
B	Automation	Nominal weight per type	Yes
C	Automation	Shared weight and cost index	No

Plan ref: to-do-list.md S11 — Altitude options and the climb rule (assumption adopted from a prior study). State the three altitude options (2,000 ft above, 2,000 ft below, 4,000 ft below) and the rule that a climb is offered only at or below FL350, citing [19] (p. 8). Say plainly that this is that study's modeling assumption, made because it did not model aircraft weight; it is not an FAA rule and not observed controller behaviour. Then explain why the rule binds cases A and B and not case C: this is the mechanism by which missing information removes an option. Still to check: that 2,000 ft steps match the levels assigned to one direction of flight ([26], paragraph not yet found).

Plan ref: to-do-list.md S11 — Baseline rule for case A. One descent clearance, citing [72]. State which aircraft descends and the tie-breaker (undecided), and whether the aircraft is cleared back up.

Plan ref: to-do-list.md S11 — Resolution options. Introduce Table 5. For each option give the instructions it costs now, the follow-up it creates, and what the decider must know to use it. Cite [34] for the instruction accounting.

Plan ref: to-do-list.md S11 — Sources for the option set and the counts (owner's note, second hand review, 2.4: "Add these to report!"). Levers: `experiment-definition.md` §4, and [19] (p. 8) for lateral and altitude changes; that study does not use speed. Altitude steps and climb rule: [19] (p. 8), with the authors' reason. Instruction counts: [34] (pp. 72–74). Climbs used less than descents: [72] (6 of 36). Speed limited to ±0.04 Mach and built to fail: the author's own figure, to be stated as experience.

Plan ref: to-do-list.md S11 — The baseline is not hobbled (owner's note, second hand review, 3.2). In the later variation at or below FL350 the baseline may climb and does not, because its rule is one descent. Say that this matches the observed preference for descents ([72]).

Plan ref: to-do-list.md S11 — What the system column is (owner's notes, second hand review, 5.5 and 6.1). State that it is a measure of the airspace as a whole system of systems, define it by its cost statement (total fuel, total CO2, total airline cost, workload items, largest cost to any one aircraft), and refer back to that statement wherever the column is discussed.

Plan ref: to-do-list.md S11 — How each option is scored, one statement per viewpoint. Aircraft: extra fuel plus cost index times extra time ([68], Eq. 1–3). Airline: the sum over its aircraft plus a step cost when a flight misses a connection, plus CO2 converted to dollars. Sector: clearances issued, effort to devise them, and residual risk, reported as three separate items. System: totals stated side by side, with no weighted sum (D-013). Say why no weighted sum: nobody in the model owns the weights.

Plan ref: to-do-list.md S11 — Emissions. State the conversion from fuel to CO2 and its source (not yet held). State how CO2 becomes dollars at the airline level and why that route was chosen (credit price, passenger value, or both; undecided), and why the system level keeps CO2 in kg.

Plan ref: to-do-list.md S11, S12 — Truth model (decided, D-017). OpenAP (open aircraft performance model, TU Delft) scores every option: fuel flow, thrust and drag for the B737-9 at the aircraft's actual mass. State the three selectable fidelities (conceptual: feasibility and instruction counts only; nominal mass; actual mass) and that the decider's case (A to D) and the truth model's fidelity are independent settings. State the limitation: OpenAP has no drag polar for the 737 MAX 9 and borrows the MAX 8's. Cite OpenAP (Sun, Hoekstra and Ellerbroek 2020) once registered. The earlier kinematic model (`vehicle_dynamics.py`) is not used by the experiment.

2.5 Scoring Model Calculations

Plan ref: to-do-list.md S11, S12 — PLACEHOLDER, AI-drafted 2026-10-11. This subsection states what `simulation/experiment_scoring.py` and `simulation/b739_envelope.py` calculate. Check each equation against the code, confirm the two cited sources by reading them (the OpenAP paper and Bartel and Young are cited here from the package source code), then rewrite in the author's words. The limitations in Section 2.5.7 are findings from writing this up and need a decision each.

2.5.1 Equations of motion

Each aircraft is a point mass in steady flight in still air in the International Standard Atmosphere. OpenAP [87] uses the point-mass equations along and normal to the flight path,

$$T - D - mg\sin\gamma = m\frac{dV}{dt}, \tag{1}$$

$$L = mg\cos\gamma, \tag{2}$$

where T is total thrust, D drag, L lift, m mass, V true airspeed and γ the flight path angle, with $\tan\gamma = \dot{h}/V$ for a vertical speed $\dot{h}$. The experiment sets $dV/dt = 0$ everywhere: every aircraft holds its Mach number. In level cruise $\gamma = 0$, so $T = D$ and $L = mg$.

True airspeed comes from the Mach number M and the standard temperature $\Theta(h)$ at the altitude h:

$$V = M\sqrt{\kappa R\,\Theta(h)}, \qquad \Theta(h) = \begin{cases} 288.15 - 0.0019812\,h & h < 36{,}089 \text{ ft} \\ 216.65 & \text{otherwise} \end{cases} \tag{3}$$

with $\kappa = 1.4$, $R = 287.05$ J/(kg K), h in feet and Θ in kelvin.

2.5.2 Drag, thrust and fuel flow (OpenAP)

Drag. OpenAP computes clean-configuration drag from a two-term drag polar:

$$C_L = \frac{mg\cos\gamma}{\frac{1}{2}\rho V^2 S}, \qquad C_D = C_{D0} + k\,C_L^2, \qquad D = \tfrac{1}{2}\rho V^2 S\,C_D. \tag{4}$$

The values used are in Table 4. This is where aircraft mass enters: a heavier aircraft needs a higher C_L and its drag rises with C_L^2.

Assumption: no compressibility drag. Eq. 4 has no Mach-number term, so the extra drag from shock waves on the wing is left out. The FAA circular classes flight below Mach 0.75 as subsonic and Mach 0.75 to 1.20 as transonic [22] (para. 1-4e), so the scenario's Mach 0.80 is at the low end of the transonic range. OpenAP has an optional wave-drag term, off by default and marked experimental by its authors. Switching it on is the only estimate of the size of this assumption held so far: at the scenario's weights and levels, drag rises by 2 to 4 percent at Mach 0.80 and by less than 1 percent at Mach 0.76. Fuel flow in level cruise moves by about the same few percent. The rate of climb left (Eq. 8) moves much more, because it is a small difference between thrust and drag; see Section 2.5.7.

Maximum thrust. Above 30,000 ft OpenAP uses the high-altitude segment of the two-shaft turbofan model of Bartel and Young [3], which scales a reference thrust at the top of climb by pressure and Mach number:

$$T_{\max} = T_{cr}\left[d\,\ln\frac{p}{p_{cr}} + \left(\frac{M}{M_{cr}}\right)^{-0.11}\right], \qquad d = -0.4204\,\frac{M}{M_{cr}} + 1.0824, \tag{5}$$

where p is the ambient pressure, and p_{cr} and M_{cr} are the pressure and Mach number at the type's reference cruise condition. The package returns the same value for its cruise and climb thrust at these altitudes, so there is one thrust limit, not two.

Fuel flow. OpenAP finds the thrust the manoeuvre needs from Eq. 1,

$$T_{req} = D + mg\sin\gamma, \tag{6}$$

and converts it to fuel flow with an empirical curve of thrust ratio, scaled by the engine's take-off fuel flow from the ICAO engine emissions databank:

$$\dot{m}_f = n\,s\left[c_1 - \exp\!\left(-c_2\left(x\,e^{c_3 x} - \frac{\ln c_1}{c_2}\right)\right)\right], \qquad x = \frac{T_{req}}{n\,T_0}, \tag{7}$$

where n is the number of engines, T_0 the sea-level static thrust of one engine, s the engine's take-off fuel flow, and x is limited smoothly to between 0.03 and 1.2. In level cruise $T_{req} = D$, so fuel flow depends on mass only through the drag of Eq. 4.

Table 4: PLACEHOLDER – values OpenAP 2.6.2 uses for the B737-9 (type code B39M), read from the installed package on 2026-10-11.

Quantity	Value	Note
Wing area S	124.6 m^2	type data
Zero-lift drag coefficient C_{D0}	0.020	737 MAX 8 polar, borrowed
Induced drag factor k	0.042	737 MAX 8 polar, borrowed
Engines n, type	2, LEAP-1B	type data
Static thrust per engine T_0	111,300 N	engine data
Reference thrust T_{cr}	46,300 N (both engines)	$0.2\,T_0 + 890$ N per engine
Reference cruise M_{cr}, altitude	0.79, 11,000 m	type data
Take-off fuel flow s	0.877 kg/s per engine	ICAO databank
Fuel curve c_1, c_2, c_3	0.9376, 1.9768, 1.3955	package default curve, not fitted to this type

2.5.3 Whether an aircraft can be cleared to a level

The rate of climb an aircraft has left in level flight at a mass, Mach number and altitude is its excess thrust times its speed, divided by its weight:

$$RC = \frac{(T_{\max} - D)\,V}{mg}. \tag{8}$$

This is Eq. 1 with $dV/dt = 0$, solved for $\dot{h} = V\sin\gamma$. A level is available to an aircraft only if

$$RC(m, M, h) \geq RC_{\min}. \tag{9}$$

$RC = 0$ is the level above which the aircraft cannot hold altitude. $RC = 100$ ft/min is the definition of the service ceiling [22]. The experiment uses $RC_{\min} = 150$ ft/min, the margin the author has seen airlines keep for clear air turbulence and other unknowns (industry experience, no document held; D-023). The envelope figure (Figure 2) draws all three lines for the heavy aircraft.

The low-speed side of that figure is not from OpenAP. It is 1.3 times the speed at which lift equals weight at an assumed maximum lift coefficient $C_{L\max} = 1.45$:

$$V_{\min} = 1.3\sqrt{\frac{2mg}{\rho S\,C_{L\max}}}. \tag{10}$$

The high-speed side is the lower of the maximum operating Mach number (0.82) and the maximum operating calibrated airspeed (340 kt).

Which mass goes into Eq. 9 is the experiment. Cases A and B do not know the mass, so they do not evaluate it: they offer a climb only at or below FL350. Cases C and D evaluate it at the aircraft's actual mass. The scoring then checks every chosen option again at the actual mass; an option the aircraft cannot fly is reported as "unable".

2.5.4 Cost of one manoeuvre

Every cost is a difference from flying on undisturbed. Write $\dot{m}_f(h, \dot{h})$ for the fuel flow of Eq. 7 at the aircraft's mass and Mach number, h_0 for the present level, h_1 for the new one, t_h for the time the manoeuvre is held (15 min) and $\dot{m}_{f0} = \dot{m}_f(h_0, 0)$.

Altitude change. A descent is flown at 1,500 ft/min. A climb is flown at maximum thrust, at the rate that thrust allows: Eq. 8 evaluated at the mid-altitude, and no more than 1,500 ft/min (D-024):

$$\dot{h} = \begin{cases} -1{,}500 \text{ ft/min} & \text{descent} \\ \min\Big(1{,}500,\ RC\Big(m, M, \frac{h_0+h_1}{2}\Big)\Big) & \text{climb} \end{cases} \qquad t_c = \frac{|h_1 - h_0|}{|\dot{h}|}. \tag{11}$$

At the scenario's weights and levels the climb rate is 250 to 500 ft/min, so a 2,000 ft climb takes four to eight minutes. Fuel flow during the change is evaluated at the mid-altitude:

$$\Delta m_f = \underbrace{\Big[\dot{m}_f\Big(\tfrac{h_0+h_1}{2}, \dot{h}\Big) - \dot{m}_{f0}\Big]\, t_c}_{\text{the level change}} + \underbrace{[\dot{m}_f(h_1, 0) - \dot{m}_{f0}]\, t_h}_{\text{the hold at the new level}} + \underbrace{\Big[\dot{m}_f\Big(\tfrac{h_0+h_1}{2}, \dot{h}_b\Big) - \dot{m}_{f0}\Big]\, t_b}_{\text{climb back, after a descent only}}. \tag{12}$$

Here $\dot{h}_b$ and t_b are the rate and time of the climb back to h_0, from Eq. 11. The time difference is the same ground distance flown at a different true airspeed,

$$\Delta t = t_h \left(\frac{V(h_0)}{V(h_1)} - 1 \right), \tag{13}$$

which is zero between levels above 36,089 ft, where the standard temperature is constant. The return to the original level after a climb is counted as an instruction but given no fuel term.

Vector. A pass-behind vector adds a fixed track distance d_x (6 NM) at the cruise speed and fuel flow:

$$\Delta t = \frac{d_x}{V(h_0)}, \qquad \Delta m_f = \dot{m}_{f0}\, \Delta t. \tag{14}$$

Slow-down (the in-trail aircraft in option O9). The aircraft flies at $M - \Delta M$ the distance it would have covered in t_s minutes at M:

$$t'_s = t_s \frac{V(M)}{V(M - \Delta M)}, \qquad \Delta t = t'_s - t_s, \qquad \Delta m_f = \dot{m}_f(M - \Delta M)\, t'_s - \dot{m}_{f0}\, t_s. \tag{15}$$

Speed change on the crossing pair. Not calculated. It is marked unable to separate the pair by construction (D-014).

2.5.5 From manoeuvre cost to the outcome table

For aircraft i the cost of an option is extra fuel plus the cost index times extra time, in kilograms of fuel, the form used in [68]:

$$c_i = \Delta m_{f,i} + CI_i \, \Delta t_i. \tag{16}$$

Each case chooses, among the options it is allowed to offer, the one with the lowest estimate

$$J = \sum_i \left(\Delta m_{f,i}(\hat{m}_i) + \widehat{CI}_i \, \Delta t_i(\hat{m}_i) \right), \tag{17}$$

where the hats mark what the case believes. Case A does not use Eq. 17; it applies its rule (one descent). Case B uses the type's nominal mass for every aircraft and values time at 1 kg/min. Cases C and D use each aircraft's actual mass and cost index. Case D also rejects an option that pushes a flight with connections past the connection threshold. Whatever a case chooses is then scored at the actual masses, and that score is what the tables report.

The other columns follow from Eq. 12 to 16. Carbon dioxide is $3.16 \, \Delta m_f$ [48]. An airline's cost in dollars is the sum over its aircraft of $c_i \, p_f + 3.16 \, \Delta m_{f,i} \, p_c$, with p_f the fuel price and p_c the carbon price. System fuel is $\sum_i \Delta m_{f,i}$. Instructions are counted, now and later, and are not converted to anything.

2.5.6 Worked example: option O4, the light aircraft climbs 2,000 ft

The light aircraft is at 62,980 kg, FL380, Mach 0.80. Eq. 8 gives 322 ft/min left at FL400, above 150, so the level is available to a case that knows the mass. Eq. 11 gives a climb rate of 409 ft/min at FL390, so the climb takes 2,000/409 min, or 294 s. Fuel flows from Eq. 7: 0.5572 kg/s at FL380 (2,006 kg/h), 0.5465 kg/s at FL400 (1,968 kg/h), and 0.6321 kg/s in the climb. Eq. 12 gives

$$\Delta m_f = (0.6321 - 0.5572)(294) + (0.5465 - 0.5572)(900) = 22.0 - 9.6 = 12.4 \text{ kg},$$

and $\Delta t = 0$ because both levels are above 36,089 ft. This is the 12.4 kg in the options table.

2.5.7 Limitations found in writing this up

1. **No compressibility drag, and the climb left is sensitive to it.** With OpenAP's experimental wave-drag term switched on, drag at Mach 0.80 is 2 to 4 percent higher. That is 1.3 kN for the heavy aircraft at FL380, against 2.7 kN of excess thrust, so its climb left falls from 181 to 93 ft/min, under the 100 ft/min of the service ceiling. The light aircraft at FL400 falls from 322 to 239 ft/min and still clears 150. The weights in the scenario were set with the term off. Not decided: whether to switch the term on and set the weights again.

2. **One climb rate per climb.** The rate is taken at the mid-altitude and held; the climb is not integrated. A climb is not checked against the time left before the crossing (12 min); the longest in the base scenario is the heavy aircraft's 4,000 ft climb back after option O8, about 12 min, which happens after the conflict.

3. **The fuel curve is the package default.** OpenAP has no fuel model fitted to the 737 MAX 9 or MAX 8; Eq. 7 uses its default coefficients scaled by the LEAP-1B take-off fuel flow.

4. **The drag polar is the 737 MAX 8's** (D-017).

5. **Mass is constant.** Fuel burned during the manoeuvre does not reduce the mass.

6. **The low-speed limit rests on an assumed** $C_{L\,\max}$ and is a stall margin, not a buffet boundary.

7. **Standard atmosphere, no wind**; time differences come only from true airspeed and added track distance.

Plan ref: to-do-list.md S11 — Abstractions made and why. One short table: each simplification, the reason, and the risk. A draft list is in `experiment-options-and-scoring.md` §6.

3 Results and Discussion

REQUIRED FOR INTERIM REPORT #2 (due 2026-10-25) — Results and Discussion (preliminary). For Interim Report #2 this section needs: (1) the architecture as it stands (context diagram, aircraft IBD, the five improved requirements as the measured gap), with the figures already in Appendix A.3; (2) the two hand checks from `experiment-definition.md` §9 written up as preliminary results; (3) the first OpenAP numbers (Figure 1: fuel flow against altitude for the heavy and light aircraft) and the baseline row of Table 6 if the scoring run is done; (4) one paragraph on what is not yet done. Mark the section "preliminary" in its heading or first line, as the course asks.

Plan ref: to-do-list.md S5 — Present the nominal-flight ConOps (strategic/commercial planning through day-of-operations, turnaround, flight execution, and postflight/continuation) as the operational backbone the architecture models. Summarize; put the full activity breakdown in Appendix A.1.

Plan ref: to-do-list.md S8 — Present the stakeholder objective/cost/value ontology: what "optimal" means per stakeholder (airline, passenger, ATC/ANSP, airport, flight crew, environmental/societal, military), and the classification of each objective as hard constraint, optimization objective, cost/penalty, MOE, or MOP. Highlight conflicting vs. aligned objectives and objectives with externalized costs.

Plan ref: to-do-list.md S9 — Present the myopic-optimization analysis: the definition of "local optimum" in the SoS context, candidate SoS measures of value, and the documented conflicts (decision maker, decision variable, local objective/constraints, affected stakeholders, externalized costs, SoS consequence). See the flagged scope note below.

Scope note (flag, not a TODO to resolve silently): Project_To-Do List.md S9 catalogs eight candidate local-vs-SoS conflicts, each with eight documented attributes (64 cells). A 15–40 page report body cannot carry all eight in full; plan to present 2–4 illustrative conflicts in the main body (favoring whichever one becomes the S11–S14 demonstration scenario) and move the remaining catalog entries to Appendix A.4.

Plan ref: to-do-list.md S10 — Present the SysML architecture: context diagram, package structure, stakeholder model, BDDs, IBDs, information-object model, interface/item-flow model, nominal-flight activity diagram, responsibility swimlanes, and requirements model with traceability. Embed the most illustrative diagrams here; move the full diagram set to Appendix A.3.

Plan ref: to-do-list.md S12, S13 — Present the optimization/simulation results: the baseline scenario, single-stakeholder vs. balanced/SoS optimization outcomes, and how simulation entities/decisions/objectives/constraints trace back to SysML blocks, properties, goals, and requirements. Show how the architecture exposes impacts a standalone optimizer would omit (S13's key point).

Scope note: Project_To-Do List.md S12 as written (Pareto fronts, full objective-weight sweeps, sensitivity analysis, tipping-point identification across the whole experiment matrix) reads as a full optimization research study layered on top of the architecture work. Per this project's own working assumption ("not an optimization paper"; optimization is a capability inside the architecture, not the whole subject) and the 15–40 page report constraint, consider scoping S12's demonstration to one representative scenario with a single weight sweep and one Pareto-style comparison, rather than the full matrix – see the note added to `knowledge/questions/open-questions.md` under "Optimization study scope (S11)."

3.1 Experiment Results

Table 5: PLACEHOLDER – resolution options. A dash means no instruction. Full version with instruction counts in Appendix A.5.

Option	AC1	AC2	AC3	AC4
O1	Descend 2,000 ft	–	–	–
O2	–	Descend 2,000 ft	–	–
O3	Climb 2,000 ft	–	–	–
O4	–	Climb 2,000 ft	–	–
O5	Turn behind AC2	–	–	–
O6	–	Turn behind AC1	–	–
O7	Reduce speed	–	–	–
O8	Descend 4,000 ft	–	–	–
O9	Descend 2,000 ft	–	Slow or turn	–

Table 6: PLACEHOLDER – outcome of each case at four levels, as a difference from case A. No simulation has been run; cells are empty.

Case	Option chosen	Aircraft (largest cost to any one)	Airline (each airline's total)	Sector (clearances / effort / risk)	System (total fuel, total CO2, total cost)
A		reference	reference	reference	reference
B					
C					

Plan ref: to-do-list.md S12 — Which option each case chose, and why. Walk Table 6 row by row. For case A, name the rule that produced the choice. For B and C, name the cost that decided it.

Plan ref: to-do-list.md S12 — Did the climb rule matter? With the crossing pair above FL350, cases A and B could not offer a climb and case C could. Say whether case C used one, for which aircraft, and what it saved. If C did not climb either, say so: that is a result too.

Plan ref: to-do-list.md S12 — B minus A, and C minus B. The first is what automation is worth; the second is what the information exchange is worth. Compare the size of each with the prior figure of tens of gallons per flight ([19], p. 12) and say whether the difference is larger than the truth model's uncertainty.

Plan ref: to-do-list.md S9, S12 — Does the best option change with the level? Read across each row of Table 6. Name any option that is better for one aircraft or one airline and worse for the sector or the system, and who pays (the largest cost to any one aircraft). This is the sub-system versus system discussion.

Plan ref: to-do-list.md S12 — Airline X's two flights. Discuss the option that touches both of one airline's aircraft: what the airline would choose between its late flight and its early one, and that the controller cannot see that preference. Connect to the dispatcher priorities (schedule integrity and connections rated at or above fuel; source of that table still to confirm).

Plan ref: to-do-list.md S12 — Workload is three things. Discuss clearances issued, effort to devise, and residual risk separately. Say where a "more workload" option on one of them is "less" on another.

Plan ref: to-do-list.md S12 — When would the controller say no? (owner's note, second hand review, 5.2). Discuss whether refusals are more likely in morning rushes at hub airports or near holidays, using the sector-loading rule in [19] (p. 10). A hypothesis; needs a source.

Plan ref: to-do-list.md S12 — Is schedule the right priority? (owner's note, second hand review, 4.11). Contrast a carbon price, which makes fuel weigh more against time, with controllers' usual practice of giving schedule the priority.

Plan ref: to-do-list.md S12 — Figure: the options on a Pareto front (owner's note, second hand review, 5.4), if that column is kept. Also needed: a figure of the scenario itself (note 1.1).

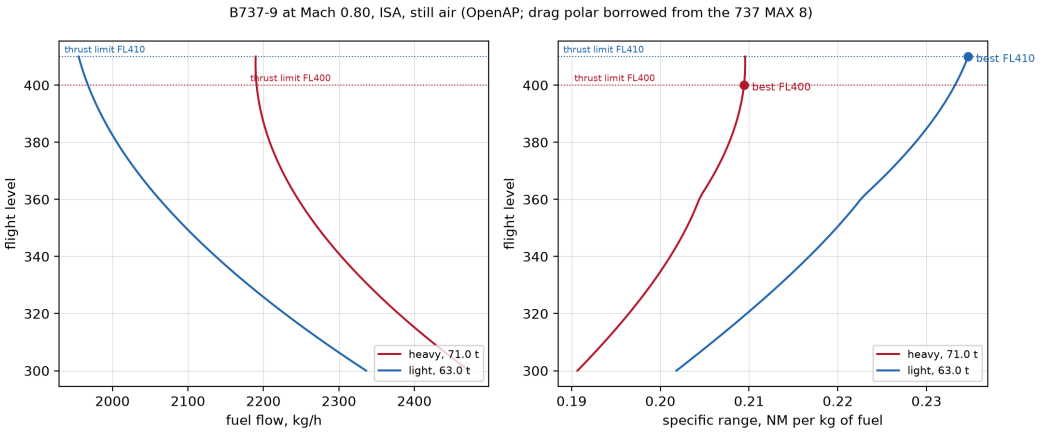


Figure 1: PLACEHOLDER – fuel flow and specific range against altitude at Mach 0.80 for the heavy and light B737-9 (OpenAP, D-017). The mechanism the experiment rests on: the light aircraft's optimum sits higher and it can climb where the heavy one cannot.

Plan ref: to-do-list.md Track A — Figure slot: sequence diagram of case C's conflict resolution (controller, ATC decision support, airline operations center, FMS, crew), rendered from the model once drafted (turn 3).

Plan ref: to-do-list.md S12 — The case where nothing changes. Report the variation with similar weights and cost indexes, where case C picks the same option as case A, to show the comparison is not rigged.

Plan ref: to-do-list.md S13 — Trace back to the model. For each column of Table 6, name the part in the context diagram it belongs to (aircraft and crew; airline operations center; ATC sector). Say what the system column maps to, since no single part owns it. Name the exchange that case C adds (weight and cost index to ATC) and where it sits in the model.

Plan ref: to-do-list.md S12, S15 — Limits. Level crossing conflicts are a minority (36 of 256, [72]); one invented sector; a static snapshot; the climb rule is one threshold for every type; airlines could misreport what they share (cite the 1 to 3 percent figure once that source is registered).

4 Recommendations

Plan ref: to-do-list.md S14, S15 (S16 bullet added this session) — Restate the main findings from Section 3 and provide recommendations for moving forward. Include the alternatives considered (e.g., alternative decompositions from S6, alternative stakeholder-weighting schemes from S11) and the criteria used to select the recommended approach. Suggest additional areas of study (candidate: architecture extensions, additional off-nominal scenarios from S15, unexplored myopic-optimization conflicts from S9's catalog).

NOTE: Project_To-Do List.md did not previously have a distinct "Recommendations" deliverable separate from "Conclusions and future work" – added to S16 this session so this section has an explicit home in the plan.

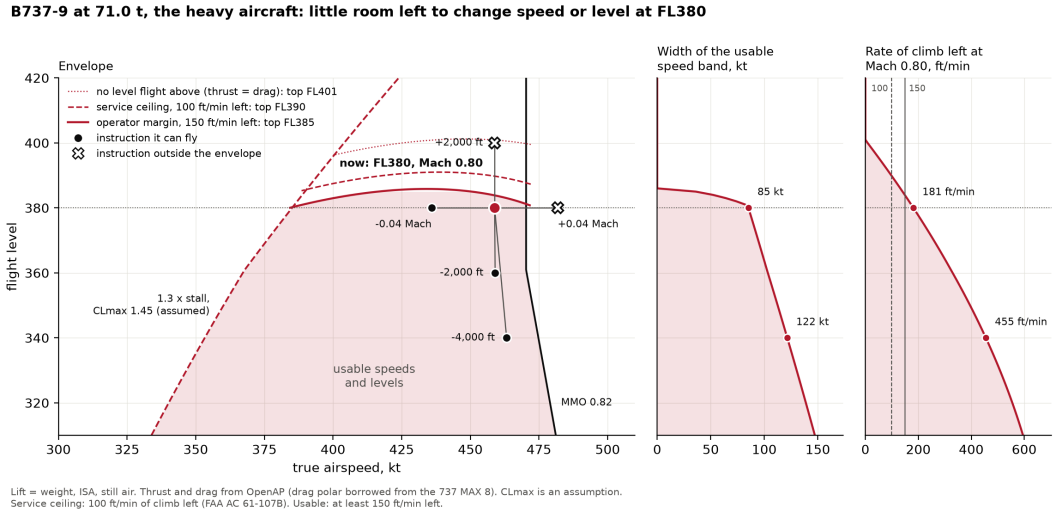


Figure 2: PLACEHOLDER – cruise envelope of the heavy B737-9 (71.0 t) assuming lift equals weight: low-speed boundary at an assumed clean maximum lift coefficient, maximum operating Mach, and three lines of constant rate of climb left: 0 ft/min (thrust equals drag), 100 ft/min (the service ceiling, [22]) and 150 ft/min (the operator margin the scenario uses, author's industry knowledge), with the aircraft's present level and Mach and each altitude and speed instruction the scenario offers it marked as inside or outside the region with 150 ft/min left. Right: width of the usable speed band and rate of climb left at Mach 0.80, by flight level (OpenAP, D-017).

5 Impact/Financial Benefits

Plan ref: to-do-list.md S9, S12 (S16 bullet added this session) — Quantify the impact of the reference architecture and its optimization-demonstration capability. Translate the S9 conflict analysis and S12 demonstration results into concrete operational terms wherever possible (e.g., delay-minutes, fuel burn, CO2, gate/runway utilization) and into US-dollar terms where a credible conversion exists (e.g., published cost-per-delay-minute figures from the literature review, Section 1.2). Also discuss the ethical context and any unintended consequences of the recommendations (equity-minded engineering: whose costs get externalized, and onto whom – this connects directly to S8's "objectives whose costs are externalized onto another stakeholder" and S9's conflict catalog).

Scope flag: the 503 template expects a quantified financial-benefit statement (the template's sample statements are dollar-savings claims from industry DMAIC-style projects). This capstone is an architecture/SoS project, not a cost-reduction project with a "before" baseline in dollars, so a literal dollar-savings figure may not be available or meaningful. Project_To-Do List.md did not previously have a task that produces a quantified/monetized impact figure – an explicit "Impact / quantified benefits" bullet was added to S16 this session so at least one deliverable targets this section directly. Recommend framing impact primarily in operational/systems terms (delay, fuel, emissions, predictability, tradeoff-surface shape) with a dollar estimate offered as an illustrative, clearly-caveated secondary figure rather than the headline metric.

5.1 Emissions: What They Mean to the Airline and to Society

Plan ref: to-do-list.md S9, S12 — Airline level: put CO2 into dollars. Take the fuel difference between cases from Table 6, convert to kg of CO2, then to dollars. State the route used and its source. Candidates: (1) the price of a carbon credit or allowance, if a scheme applies to the flight; (2) passenger value, meaning revenue gained or lost because some passengers choose the flight listed with lower emissions. Make the point that fuel is already counted once as fuel cost and that pricing its CO2 counts the same kilograms again, on purpose. SOURCES NOT YET HELD.

Plan ref: to-do-list.md S9, S12 — Passenger value. Discuss how passengers see a flight's emissions when booking, and whether that reaches the airline as demand. The author's own practice of choosing lower-emission listings is the motivation, not the evidence: cite a study. Note the gap between one decision in one sector and the label a passenger saw at booking; the link is the airline's average over many flights.

Plan ref: to-do-list.md S9, S12 — System-of-systems level: keep CO2 in kg. Report total CO2 across all four aircraft for each case. Discuss what it means for society and for how passengers perceive air travel. A social cost of carbon may be quoted for scale, labelled as a cost nobody in the scenario pays. Connect to S8's objectives whose costs are externalized onto another stakeholder.

Plan ref: to-do-list.md S12 — Scale it honestly. One conflict in one sector is a small number. If a per-year or fleet-wide figure is offered, state the multiplier and its source (for example, how often level crossing conflicts occur), and label it illustrative.

Plan ref: to-do-list.md S9 — Who pays. For the option the system prefers, name the aircraft or airline that is worse off and by how much. This is the equity discussion the template asks for.

6 Summary/Conclusions

Plan ref: to-do-list.md S15, S16 — Briefly restate the problem, approach, and findings/ recommendations, including potential benefits in relative terms if absolute figures are sensitive or incomplete (e.g., "reduces network-level delay propagation by an illustrative X% under the demonstrated weighting" rather than an unsupported absolute claim). Cover: known architecture omissions, limitations of the selected decomposition (S6), limitations of the optimization demonstration (S12 scope note, Section 3), and opportunities for future expansion (e.g., additional off-nominal scenarios, unexplored S9 conflicts, deeper crew/maintenance-routing integration).

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A Appendices

Plan ref: to-do-list.md S5, S7, S10, S11 (general) — Use appendices for material too extensive for the main body or of interest to only some readers. Label each lettered (Appendix A, B, ...), and ensure every appendix is referenced from somewhere in the main body.

A.1 Full Nominal-Flight ConOps

Plan ref: to-do-list.md S5 — Full strategic-planning through postflight/continuation activity breakdown (see main-body summary, Section 3).

A.2 Full Stakeholder Register and RACCI Matrix

Plan ref: to-do-list.md S7 — Complete stakeholder register (needs, goals, responsibilities, authorities, resources, constraints, information needs, measures of value) and the full RACCI matrix around operational decisions.

A.3 Full SysML Diagram Set

Plan ref: to-do-list.md S10 — Diagrams not embedded in Section 3: full BDD/IBD set, requirements model, and complete traceability matrices (needs-¿objectives-¿requirements-¿systems-¿activities-¿information exchanges-¿aircraft behavior). The figures below are every view in the model, generated by `tools/render_sysml_diagrams.py`. Not yet modeled as views: BDDs and the requirements model.

Each figure is a view defined in `nas_package_my_views.sysml` and rendered from the SysML v2 text with Syside. The caption names the view, the `.sysml` file that holds the content, and the date of the render.

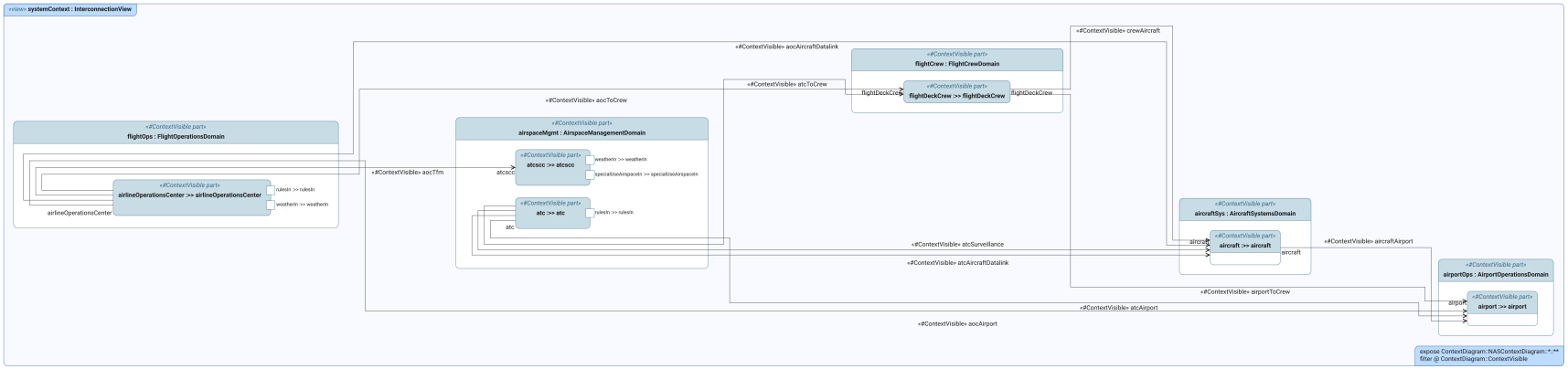


Figure 3: NAS context diagram. View `systemContext`; source `nas_context_diagram.sysml`; rendered 2026-10-10.

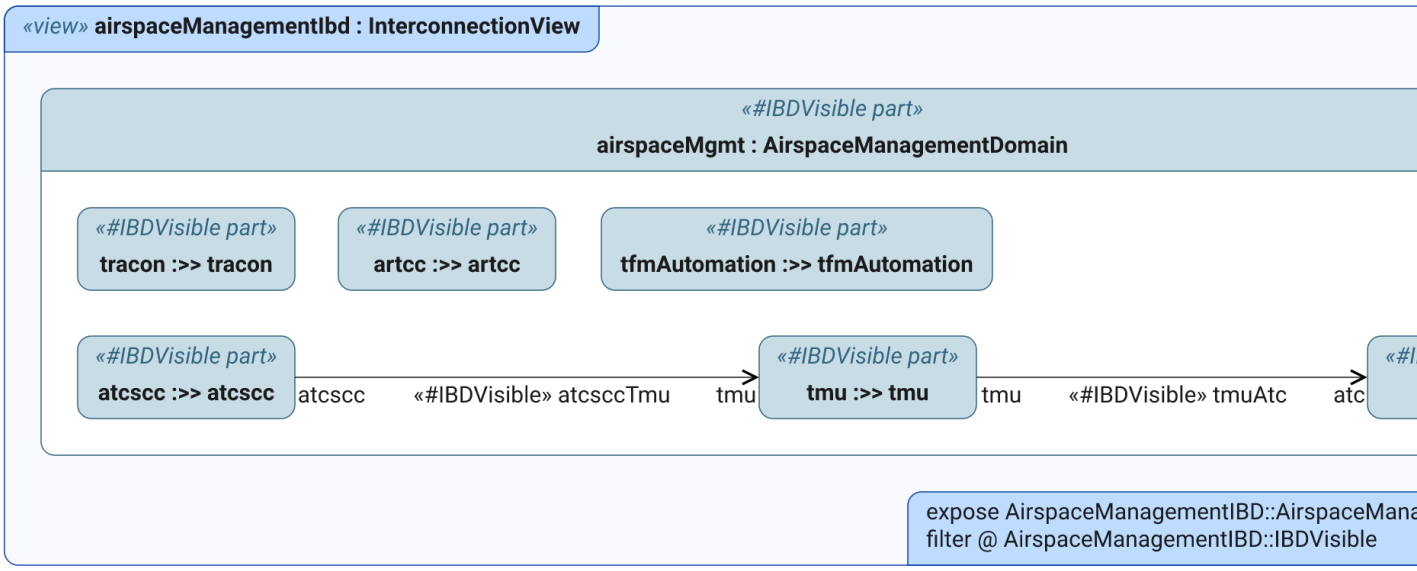


Figure 4: Airspace management internal block diagram. View `airspaceManagementIbd`; source `nas_ibd_airspac` rendered 2026-10-10.

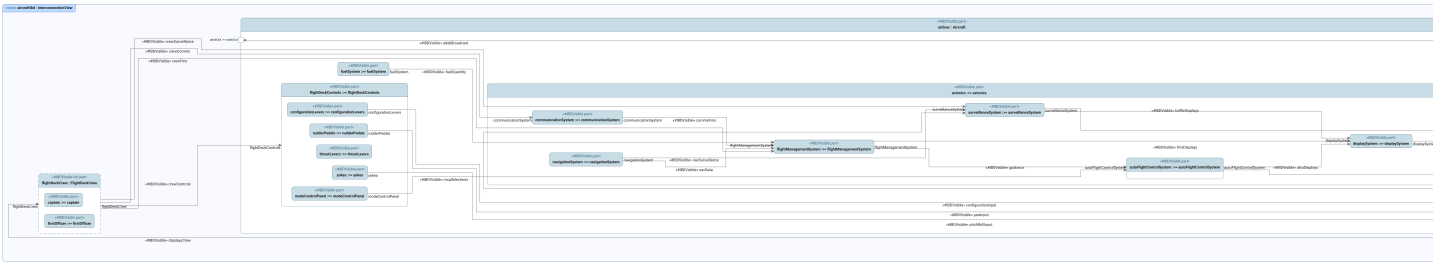


Figure 5: Aircraft internal block diagram, default level of detail. View `aircraftIbd`; source `nas_ibd_aircraft.` 10-10.

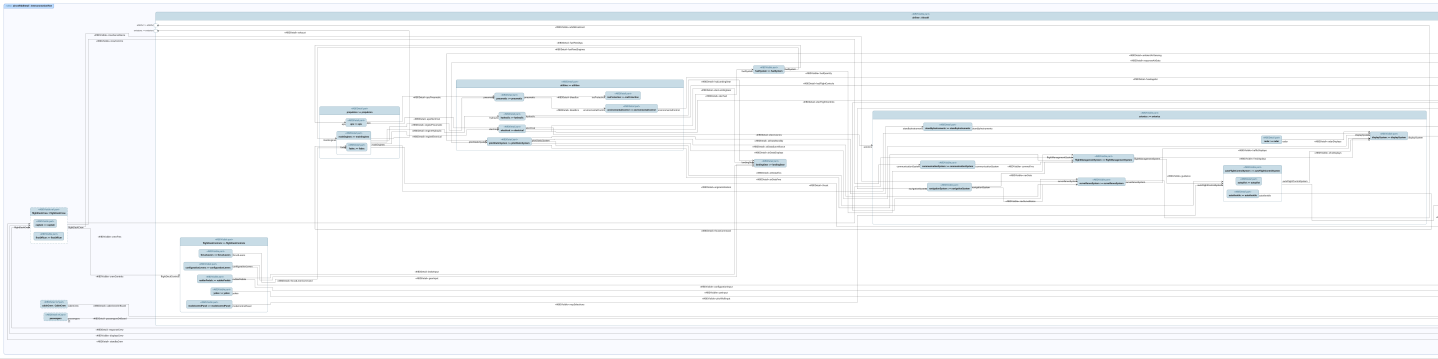


Figure 6: Aircraft internal block diagram, detailed level: adds propulsion, airframe, landing gear, utilities, and th
View `aircraftIbdDetail`; source `nas_ibd_aircraft.sysml`; rendered 2026-10-10.

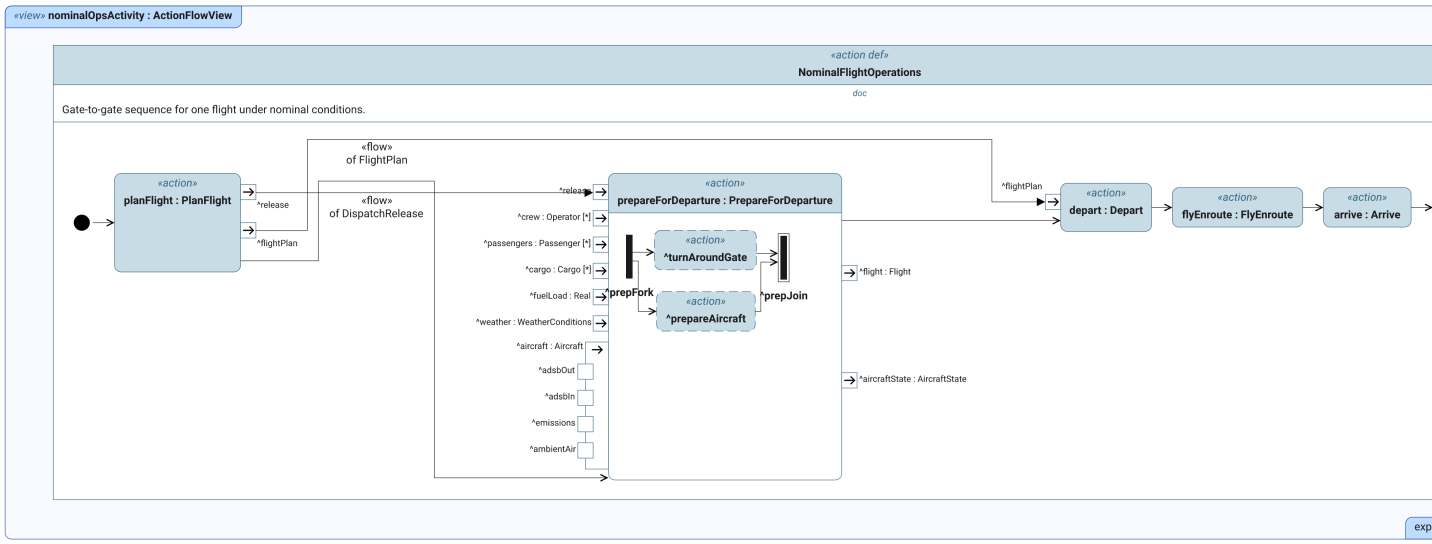


Figure 7: Nominal flight operations activity diagram (draft: no swimlanes, sub-actions are placeholders). View
source `nas_activity_diagram-nominal_ops.sysml`; rendered 2026-10-10.

A.4 Full Myopic-Optimization Conflict Catalog

Plan ref: to-do-list.md S9 — All candidate conflicts not carried into the main-body discussion or the S11–S14 demonstration scenario, each with: decision maker, decision variable, local objective/constraints, affected stakeholders, externalized costs, SoS consequence, and information needed to recognize the consequence.

A.5 Optimization Experiment Detail

Plan ref: to-do-list.md S11, S12 — Full experiment matrix, constituent-model validation notes, and any sensitivity-analysis detail not included in Section 3.

Plan ref: to-do-list.md S11 — Full options table: for each of the nine options, the maneuver per aircraft, instructions now, follow-up later, and what the decider must know to use it. Draft in `knowledge/models/experiment-options-and-scoring.md` §2.

Plan ref: to-do-list.md S11, S12 — Every option scored at four levels (the table from which each case picks one row), the per-viewpoint cost statements with their symbols defined, and the table of abstractions with reason and risk.

A.6 Generative AI Use Disclosure

REQUIRED FOR INTERIM REPORT #2 (due 2026-10-25) — Generative AI disclosure: owner to draft. Required by the course policy whenever GenAI is used; Interim Report #2 is the first submission after the policy was read (2026-09-21).

A.7 Source Register

Plan ref: to-do-list.md S1–S4 — Reference or reproduce the evidence/source register (`evidence/source-register.md`): source, publication type, NAS lifecycle phase, systems/stakeholders addressed, architectural and optimization evidence extracted, and confidence/limitations.